

Trainer Debrief — Lab 9

Data Types, Operational Definitions and the Data Collection Plan

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Build a rigorous data collection plan for your capstone (A4).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-09-data-collection-raw-extract.xlsx

The extract contains 3 missing burst-pressure values (rows MMD-4137, 4218, 4301), 2 unit errors recorded in Pa instead of kPa (MMD-4164, 4255), one impossible negative value (MMD-4193), and an inconsistent PASS/FAIL vocabulary (PASS/PASS /pass/Pass). A sound operational definition must specify the unit, the measurement instrument, the decision rule and a single allowed vocabulary.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Every capstone metric has a written operational definition that two people apply identically, stratification fields are captured at source, and the plan has survived a dry run.

Where this sits in the capstone

Re-derive data classification and operational definitions in full, then build the data collection plan that will feed every statistical test in Days 3 and 4. Bad data collected now cannot be rescued by clever statistics later. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27
Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30

If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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